UNIFIED MECHANICS · THE CORPUS

The Formal Register

Every claim numbered, every dependency drawn and shipped, every obligation graded with the reason it stands: the corpus stated so that it can be checked rather than read.

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What this volume is for. The Monograph tells the framework as one continuous argument and is meant to be read. The Observational Standard holds it to the measured record. Systems of Interpretation records where it grips and where it slides off. This volume does none of those things. It states the framework as a formal object: one notation, numbered commitments, numbered definitions, numbered results with their hypotheses and their proofs or citations, a certified numerical register, the dependency graph with its complete edge ledger, and the proof obligations with the reason each is outstanding. A reader who wants to check whether the chain is sound should be able to do it here without reading the story at all.

What this volume is not. It is not a new derivation and it introduces no result that is not already carried by the Monograph or by a paper in the corpus. Where a proof is long it is stated compactly and the full derivation is cited by paper number. Where a claim is conditional the condition is written into the statement rather than into a footnote. Nothing here is graded higher than the corpus grades it elsewhere, and if this volume and any other document disagree, the Freeze Manifest of the current printing governs.

What changed in this printing. The dependency graph is now shipped as source and certified by a script rather than asserted, and its stated counts are corrected: the Fourth Printing reported thirty-one nodes and thirty-three edges, which were not reproducible because the generating source was never included. Eleven results are added, numbered T29 to T39, carrying Papers 21 through 26. Four of the seven proof obligations are discharged outright and the remaining three are regraded. The notation promise is bounded to what it actually delivers. Axiom six is reclassified as the commitment it is.

PART ONE

Notation

The symbols below are the ones the framework's own numbered claims use. This is a bounded list and it is stated as one: it does not claim to catalogue every symbol appearing anywhere in the corpus, because the papers carry working notation of their own that is defined where it is used. Earlier printings made the wider claim and did not deliver it.

```
phi         the self-description fixed point, phi^2 = phi + 1
r           the retained fraction of one act of self-description, 1/(2 phi)
u           the outward fraction, 1 - r
W_L W_B W_M   the three weights: direct, related, held
tau         an ordered operational traversal; also proper time where marked
[tau]       the equivalence class of traversals defining a recoverable object
Lambda      the carrier lattice;  Lambda* its dual
Phi(E8)     the 240 roots of E8;  alpha beta gamma  individual roots
W(.)        the Weyl group of a root system
C           the system-preserving cluster projection (conditional expectation)
K           the decay generator;  B = ker K  the readable algebra
P_X P_T P_I   the geometry projectors: space, local clock, closure
V           the five-component cluster field
e, g        coframe and metric
D, F        covariant traversal operator and its curvature
G           the clustered gravity operator
M           the mass operator
Xi          the global carrier closure residue
lambda      the standing vacuum readout, (4 pi / sqrt3) r^240
l           the carrier-to-physical length conversion
hbar        the quantum of phase
```

On identifiers. Claims are numbered A1 to A6, D1 to D18, T1 to T39 and O1 to O7, and those numbers are the corpus-wide identifiers used by every paper. They are not re-lettered in this printing. A longer prefixed form was considered and rejected: renaming would break the cross-references in nine issued papers to solve an ambiguity that does not exist, since no other numbering in the corpus uses these forms.

PART TWO

Commitments and admissibility conditions

These are what the framework commits to. Everything numbered in Part Four descends from them and from nothing else, which Part Six verifies. The heading says commitments rather than axioms because A6 is not an axiom: it is the definition of a class of maps, and calling it an axiom misdescribed it in earlier printings.

A1 Systemhood. A system is a system: it may hold differences internally without ceasing to be one thing. AXIOM.

A2 The ground. The first distinction does not land in a waiting space; the holding of the distinction is what a where is. POSTULATE, stated in Movement Two rather than hidden.

A3 The two-sided reading. A self-description reads its mixed relation in both directions, outward against held and held against outward. POSTULATE. This single commitment fixes the factor of two in r, forces the carrier's self-duality, and forces its evenness. It is the join between the numerical branch and the carrier branch of the whole framework: see Part Six.

A4 Saturation. An accumulating single channel reaches a condition where one channel can no longer carry what it has taken on. POSTULATE. Where that condition sits is OPEN.

A5 Conservation of the readout. Nothing exposed at a closure event vanishes from the total of continuing face, outgoing record, and the ordering relation between them. POSTULATE.

A6 Admissible reduction. A system-preserving reduction is unital, idempotent, completely positive, bimodular over the readable algebra, and covariant under equivalent carrier descriptions. DEFINITION, not a choice among options: any map failing one of these five conditions deletes rather than decays. The covariance clause, together with the bimodularity clause, is what forces the reduction to be a group average; see T37.

PART THREE

Definitions

D1 Distinction. That which prevents complete sameness. Primitive.

D2 Identity. The recoverability of a distinction through admissible change.

D3 Relation. The operation through which distinct contents affect, compare or transform one another without being erased.

D4 The seven doings. Distinguish, hold, close, answer, change, order, record. The constitutive grammar.

D5 Traversal stages. Persist, distinguish, relate, propagate, resolve, return, close. The grammar of one act passing through a world of finished things. Related to D4 by the noun rule: the first stage is the first three doings completed.

D6 Object. A stable equivalence class of closed traversals. Objecthood is retrospective.

D7 The update. The two-channel step carrying (resolved, held) to (resolved plus held, resolved).

D8 Channels. The outward share u and the retained share r, with u plus r equal to one.

D9 The three weights. The terms of (u + r) squared: W_L equals u squared, W_B equals 2ur, W_M equals r squared.

D10 Crossing. One act of self-description enacted across a channel. Its price is the retained fraction r of T4; see T6 for why the price is exact.

D11 Record. A distinction that remains discriminable and recoverable under the relevant future dynamics. Two-valued: there is no partially recoverable record.

D12 Observer. Any system in which a distinction of another system becomes physically related and persistently retained. Mechanical observer: an observer with staged architecture, being a prepared ready state, a channel-defining coupling, propagation, resolution, amplification, and record production.

D13 Carrier. The structure on which a self-describing record is written, required to equal its own dual.

D14 Primitive crossing. The smallest nonzero admissible carrier displacement. In E8 these are exactly the roots, of squared norm two.

D15 Cluster projection. The reduction C from carrier observables onto the readable subalgebra, satisfying A6. Its concrete form is T31.

D16 Decay generator. K, the difference between doing nothing and applying C. Minimal form I minus C. See T14 for why the minimal form is not always the one meant.

D17 Readable algebra. B, the part of the carrier algebra the decay generator leaves alone. This is the physical theory.

D18 Geometry projectors. Mutually orthogonal P_X, P_T, P_I onto external distinction, local causal order and system closure, summing to the geometry block. The physical projector is P_X plus P_T; P_I is retained as closure data and excluded from traversable directions.

PART FOUR

Results

Each result carries its hypotheses, its grade, and either a compact proof or a citation to the paper that carries the full derivation. A result stated as CONDITIONAL names its condition in the statement itself. T29 to T32 are new in this printing and are carried by Paper 21.

T1 The seven doings admit a unique linear order. PROVED. Nothing is held before it is drawn, nothing closes before it is held, nothing answers before there is a closed thing to answer, nothing changes except by answering, nothing orders except as change follows change, nothing is recorded before it has happened. The dependency relation is a strict partial order with a unique linear extension. Monograph, Movement Two.

T2 The minimal update is unique. PROVED. Requiring an update that adds and never subtracts, carries exactly one clean copy of each channel forward, and promotes the present resolved amount to the next reserve, admits exactly one operator. Movement Four.

T3 The stable ratio is the golden fixed point. PROVED. Requiring the update to preserve the ratio between channels while changing overall size yields phi squared equals phi plus one, which has exactly one positive solution. Movement Four.

T4 The retained fraction is 1/(2 phi). PROVED given A3. The two-sided reading measures the retained side with a factor of two; u and r are then pinned with no adjustable quantity. Movement Four.

T5 The three weights exhaust the whole. PROVED. Two channels squared give four combinations; the two mixed terms are the same pair read in opposite order and count once when reading direction is irrelevant. The three terms sum to one. Movement Five.

T6 The quantised crossing. PROVED given D11. A crossing either closes, at price exactly r, or fails to traverse. Intermediate prices would require a partial closure, and D11 admits none. Exactness follows from the discreteness of closure rather than from tuning. Paper 18.

T7 The traversal price in closed form. PROVED. r cubed equals one over eight phi cubed equals root five minus two, all over eight. Movement Seven; the carrier derivation of the exponent is T10.

T8 The carrier is E8. CONDITIONAL on A3. Positive-definite even unimodular lattices exist only in ranks divisible by eight; at rank eight there is exactly one, namely E8, with 240 roots. At rank sixteen there are two and at rank twenty-four there are twenty-four, so the uniqueness is a property of the first realised rank. Movement Nine; lattice classification borrowed with its name attached.

T9 Evenness descends from the two-sided reading. PROVED given A3. A relation read outward and back counts its self-pairing twice, so carrier norms fall on the even integers. Positive-definiteness follows separately from D1, since an element pairing to zero with itself is a distinction making no difference to itself. This reduces T8 from three stipulations to one. Movement Nine.

T10 The minimal closed carrier history is triangular. PROVED. For roots alpha and beta with inner product minus one, alpha plus beta is a root; setting gamma to its negative gives a sum of zero. Immediate reversal is cancellation and encloses nothing, so three is minimal. With T6 the weight of that history is r cubed. Paper 20.

T11 The root cell area is root three. PROVED. The parallelogram spanned by two roots at that angle has area the square root of four minus one; the closed triangle is half of it. This identifies the root three carried since Movement Nine as a covolume. Paper 20.

T12 The golden ring pairing. PROVED numerically to machine precision. Projected onto the Coxeter plane the 240 roots fall into eight rings of thirty whose radii pair off, four pairs, each in the ratio phi, with residual below one part in ten to the fifteenth. Research Archive; reproducible from the Cartan matrix and the eigenvectors of the Coxeter element.

T13 The physical algebra is the kernel of the decay. PROVED given A6. System identity cannot decay, so it lies in the kernel; what the generator suppresses becomes unreadable; the readable algebra is exactly the fixed algebra of the reduction. Paper 20; Movement Ten.

T14 Minimal and general decay generators are distinct. PROVED. The minimal generator I minus C is idempotent, hence has spectrum in the two-point set, and its adjoint action has eigenvalues in minus one, zero and one. That suffices for T13 and for the closure theorem, and it does not suffice for a graded descent or for a mass spectrum, which would collapse to two values. The physical generator is any positive operator annihilating the identity whose kernel is B. Paper 20, Appendix.

T15 The local clock mode is single. PROVED. The record accumulates along one monotone order; two independent orders of accumulation would be two records and therefore two systems. Movement Eleven, from the arrow of Movement Thirteen.

T16 The closure mode is single. PROVED. There is one enacted closure whose record is inhabited; the closure mode counts enacted closures. Movement Eleven, from the one-world result of Movement Twenty-Three.

T17 The inherited form is Lorentzian. CONDITIONAL on the opposite closure roles of spatial opening and causal completion. The inherited quadratic form is P_X minus P_T, up to overall sign and the conversion factor, which is the unique minimal nondegenerate form admitting null directions. Movement Twelve.

T18 Second witness for T17. PROVED, cited. Any bijection preserving the causal order of a flat continuum of more than two dimensions is a Lorentz transformation composed with a translation and a dilation, the dilation being the unit convention. Alexandrov and Zeeman; Paper 19. Independent of T17's route and reaching the same signature.

T19 Superposition is forced. PROVED. If two actions are admissible and no physical event has distinguished one, selecting either adds information the system does not contain. The description must retain both. Movement Seventeen.

T20 Complex phase is forced. PROVED. Additive action under sequential composition, unit modulus under reversible carriage, and continuity admit exactly one family of weights, the rotations, giving the exponential of the action in units of the quantum of phase. Movement Seventeen.

T21 The Born pricing is forced. PROVED, cited. Requiring the same physical effect to carry the same probability in every complete measurement context admits exactly one pricing, the squared magnitude. Gleason, with the extensions of Busch and of Caves, Fuchs, Manne and Renes for the low-dimensional cases. Movement Eighteen.

T22 Closed carriage is unitary. PROVED, cited. A transformation preserving every transition probability is unitary or antiunitary; continuity with the identity excludes the second. Its generator exists and is self-adjoint. Wigner; Stone. Movement Eighteen.

T23 The boundary weight is the interference term. PROVED as arithmetic. Expanding the squared magnitude of a sum of two unresolved alternatives at primitive magnitudes u and r gives 0.477457514, 0.427050983 and 0.095491503, summing to one: exactly W_L, W_B and W_M. The cross-term, belonging to neither alternative alone, is interference. CONDITIONAL as physics on the primitive magnitudes being u and r. Movement Eighteen.

T24 The three weights structure the gravitational account. PROVED as algebra. The square of a defect formed from curvature weighted by u against realised area weighted by r has three terms and no others: the closed curvature record, the curvature-realisation cross-term, and the realised volume. Local gravitational dynamics reside in the cross-term. PROPOSED as physics for the identification of that cross-term with ordinary gravitational dynamics. Movement Nineteen; Paper 20.

T25 Sector classification by one operator. PROVED given T14. Carrier operations commuting with the decay preserve the clustered identity and are gauge; operations connecting distinct decay sectors are matter when they close into a cycle reproducing their readable identity; failure to commute measures the cost of persistence and is mass. Modes commuting with the decay are massless. Movement Twenty; Paper 20.

T26 Retention over replacement. PROVED given D7. A persistent self-description must join each new resolution to the record rather than overwrite it, or its later states are not records of the same system. Movement Twenty-Six.

T27 Records are single, and which record is unreadable. PROVED. A state in which both alternatives remain coherently present is not discriminable as either and is therefore not a record; singleness is the definition of one. Any internal answer to which outcome would itself be a record produced by a resolving event, so the question regresses and the selection is not readable from within the record domain. Paper 18.

T28 No recollapse while the record stands. CONDITIONAL on A5. Contraction of the causal domain toward coincidence would press resolved distinctions back into unread simultaneity, which is overwrite by geometry and is forbidden by T26. Paper 18.

T29 The spatial rank is the dimension of the surviving block. PROVED given T15, T16 and D18. Since the geometry projectors are mutually orthogonal and the clock and closure modes are each single, rank P_X equals three is identical to the statement that the geometry block has dimension five, and therefore, in a rank-eight carrier, to the statement that the decayed part has rank three. Paper 21, Section Two. New in this printing.

T30 E8 has exactly three rank-three subsystems. PROVED. Up to Weyl conjugacy they are A1+A1+A1, A2+A1 and A3, with orthogonal complements D4+A1, A5 and D5 respectively. The enumeration is exhaustive because the Weyl group is transitive on the 240 roots, so every rank-three subsystem is conjugate to one containing a fixed root. All three leave a rank-five survivor. Paper 21, Section Three. New in this printing.

T31 The decay is on A3 and the surviving symmetry is D5. CONDITIONAL on the reduction averaging over a reflection subgroup of the carrier, which is the reading of A6's covariance clause recorded as O1. Given that, A1+A1+A1 is excluded by T10 for containing no closed triangular history, and A2+A1 is excluded by T25 for being reducible and therefore two decay operators rather than one. The survivor is unique. Its reduction is averaging over W(A3), the symmetric group on four letters; its gauge algebra is so(10); its matter modules are the spinor sixteen and the vector ten; and its family index splits as one invariant direction plus three, giving three families under T25's criterion. Paper 21, Sections Four to Seven. New in this printing.

T32 A complete carrier closure exists. PROVED by explicit construction. The 240 roots sum to zero because a root system contains the negative of each root, so any traversal using each root once closes automatically. An ordering in which every step is itself a root exists, and a 240-step witness is recorded in the certificates and verified independently of the search that produced it. Uniqueness and the number of such closures remain OPEN and are not required for the vacuum readout, since every complete closure has exactly 240 crossings. Paper 21, Section Eight.

T33 The action is the square, with the three weights as its coefficients. PROVED. Taking the defect of T24 with the full carrier curvature rather than the geometry block alone gives u squared times curvature-curvature, 2ur times curvature-area, and r squared times area-area. That is MacDowell-Mansouri term for term, and the identification is overdetermined: two independent coefficient ratios both return the same cosmological constant, three r over u, which is three over root five exactly, with nothing available to tune. The projection MacDowell-Mansouri inserts by hand is the cluster projection. Papers 24 and 25.

T34 The surviving space is the three-sphere. PROVED given Perelman's theorem, borrowed with its name attached. Three spatial modes, a closure making every traversal return, and one world making the space simply connected. A closed simply-connected three-manifold is the three-sphere. Paper 24, Section Four.

T35 Hydrogen's degeneracy is n squared and its energies go as minus one over n squared. PROVED given T34 and T8. Since the carrier equals its own dual there is no position-against-momentum distinction to draw, so the framework's sphere and Fock's sphere are one sphere. Bound states are then harmonics of degree n minus one, the whole s, p, d and f shell structure follows from the harmonic count, and the Balmer ratios need no coupling. The scale, the Rydberg, needs the coupling and is not claimed. Paper 24.

T36 Chirality is the traversal direction selected by the arrow. CONDITIONAL on the bridge from a forbidden traversal composition to a field-theoretic mass term. Reversal exchanges the spinor sector with its conjugate, checked across all 128 roots; the record accumulates one way by T15 and joins rather than overwrites by T26, so only one orientation reproduces a readable identity. A mass term joining the two composes a cycle with its own reverse, whose net displacement is zero, which is the overwrite T26 forbids. The Distler-Garibaldi theorem does not apply: its first hypothesis requires an SL(2,C) subgroup with the gauge group as centraliser, and the decay group here is finite of order 24. Paper 22.

T37 The reduction is forced to be the average over the decay's own reflections. PROVED given T13 and A6, with two standard results cited: a norm-one projection onto a subalgebra is a conditional expectation, and the covariant conditional expectation onto a finite-group fixed algebra is unique and is the average. The remaining condition, that the decayed part is root-generated, follows because T13 requires an algebra rather than a subspace: no rootless rank-three subspace leaves enough roots orthogonal to it to build a rank-five reductive algebra, the best managing twenty against the forty D5 requires. This removes the condition T31 carried. Paper 26, Section Two.

T38 The Standard Model charges follow from the three-two split. PROVED given T31. Hypercharge must be block-constant and traceless, which fixes its ratio; one generation is the even exterior algebra of the five complex directions; and all sixteen states, with every hypercharge and every electric charge, match the Standard Model written out independently. The weak mixing angle at unification is three eighths. Anomaly freedom was verified by summation over the states actually built rather than cited. Paper 25.

T39 The Planck to cosmological hierarchy is r to the minus half the root count. PROVED as an identity. The two routes to the cosmological constant, one from the three weights and one from one retention per root, fix the carrier-to-physical length exactly: the ratio to the Planck length is the square root of three root three over four pi root five, times r to the minus one hundred and twentieth. The large number is not a coincidence requiring a mechanism; it is a retention over half the carrier's crossings. Paper 26, Section Three.

PART FIVE

The numerical register

Every constant the framework uses, in closed form and to twelve significant figures. No value on this list is fitted, and no value is obtained by comparison with a measurement. All are reproducible from the Reproducibility layer by a single command.

```
phi          (1 + sqrt5)/2             1.61803398875
r            1/(2 phi) = (sqrt5-1)/4   0.309016994375
u            1 - r                     0.690983005625
W_L          u^2                       0.477457514063
W_B          2ur                       0.427050983125
W_M          r^2                       0.0954915028125
             W_L + W_B + W_M           1.00000000000
r^3          (sqrt5 - 2)/8             0.0295084971875
1 - r^3      retained, one crossing    0.970491502813
(1-r^3)^3    retained, three crossings 0.914061068091
sqrt3        root cell parallelogram   1.73205080757
sqrt3 / 2    closed triangle cell      0.866025403784
```

```
r^240       global carrier residue      3.94246077122e-123
lambda      (4 pi / sqrt3) r^240        2.86033313614e-122
```

Three of these deserve a note. The traversal price is written as a surd throughout the corpus because a decimal in that position reads as something fitted, and the closed form shows what it is: three copies of one retained fraction, multiplied because the channels are crossed in series. The global residue is one retention per root over a complete closure, and as of this printing that closure is proved to exist rather than granted, which is T32. The standing vacuum readout is a computed value with a stated construction and no adjustable quantity anywhere in its map; Paper 21 Section Nine reports its agreement with the measured cosmological constant, and that agreement is offered as a test the framework passes rather than as a derivation of the measurement.

PART SIX

The dependency graph

The Monograph's governing promise is that nothing is slipped in by hand. That promise is a claim about structure and can therefore be checked rather than believed. Draw every commitment, definition, result and obligation as a node, draw an edge from each to everything it uses, and ask whether the graph contains a cycle. A cycle would mean a claim standing on itself, which is the formal shape of the failure the promise forbids.

This printing ships the graph as source. Every edge below carries the phrase from this volume's own wording that licenses it, so a reader can check each one against the text rather than taking the count on trust. The Fourth Printing stated thirty-one nodes and thirty-three edges; those figures were not reproducible, because the generating source was never included, and they are corrected here.

71 nodes, 82 dependency edges, no cycles

Two kinds of link are deliberately excluded from the graph and both exclusions are recorded in the source. A reference is a pointer for the reader, not a dependency: D10 points at T6 and D16 points at T14, and treating either as a dependency manufactures a false cycle, since T6 depends on D10 and T14 depends on D16. A witness is an independent corroboration rather than a step in the chain: T18 reaches T17's conclusion by a separate route and is not part of the derivation of it.

The load-bearing commitments, those with something depending on them, are A3, A5, A6, D1, D10, D11, D13, D14, D15, D16, D17, D18, D4, D7, D9. The remaining nine, A1, A2, A4, D12, D2, D3, D5, D6, D8, are stated but carry nothing in the numbered chain. That is not automatically a fault, since several are framing rather than machinery, but it is recorded here rather than left to be discovered. A6 and A3 carry the most.

The longest chain in the corpus is 8 steps. There is no deep sequence in which an error could hide: the structure is shallow and wide, which is what a framework that unfolds from a seed looks like, and it is the checkable form of the claim that the whole thing comes from one line.

The edge ledger

Read each line as: the claim on the left uses the claim on the right, for the reason quoted.

```
D10  <- T4    Its price is the retained fraction r
T1   <- D4    The seven doings admit a unique linear order
T2   <- D7    The minimal update is unique
T3   <- T2    Requiring the update to preserve the ratio between channels
```

```
T4   <- A3    PROVED given A3
T4   <- T3    u and r are then pinned
T5   <- D9    the terms of (u + r) squared
T5   <- T4    Two channels squared give four combinations
T6   <- D10   A crossing either closes, at price exactly r
T6   <- D11   PROVED given D11
T7   <- T4    r cubed equals one over eight phi cubed
T7   <- T6    three copies of one retained fraction
T7   <- T10   the carrier derivation of the exponent is T10
T8   <- A3    CONDITIONAL on A3
T8   <- D13   required to equal its own dual
T8   <- T9    This reduces T8 from three stipulations to one
T9   <- A3    PROVED given A3
T9   <- D1    an element pairing to zero with itself is a distinction making no
T10  <- D14   In E8 these are exactly the roots
T10  <- T6    With T6 the weight of that history is r cubed
T10  <- T8    For roots alpha and beta
T11  <- T10   the closed triangle is half of it
T12  <- T8    Projected onto the Coxeter plane the 240 roots
T13  <- A6    PROVED given A6
T13  <- D15   the fixed algebra of the reduction
T13  <- D17   the readable algebra is exactly the fixed algebra
T14  <- A6    the definition, not a choice among options
T14  <- D16   The minimal generator I minus C is idempotent
T15  <- D11   two independent orders of accumulation would be two records
T15  <- T26   from the arrow of Movement Thirteen
T16  <- T27   from the one-world result of Movement Twenty-Three
T17  <- D18   The inherited quadratic form is P_X minus P_T
T17  <- T15   the opposite closure roles of spatial opening and causal completio
T17  <- T16   the opposite closure roles of spatial opening and causal completio
T19  <- D11   no physical event has distinguished one
T20  <- T19   Additive action under sequential composition
T21  <- T20   the same physical effect to carry the same probability
T22  <- T20   A transformation preserving every transition probability
T23  <- T5    exactly W_L, W_B and W_M
T23  <- T21   Expanding the squared magnitude of a sum
T24  <- T5    has three terms and no others
T25  <- T14   PROVED given T14
T26  <- D7    PROVED given D7
T27  <- D11   is not discriminable as either and is therefore not a record
T28  <- A5    CONDITIONAL on A5
T28  <- T26   which is overwrite by geometry and is forbidden by T26
T29  <- D18   P_X, P_T, P_I sum to the geometry block
T29  <- T15   rank P_T = 1
T29  <- T16   rank P_I = 1
T30  <- T8    rank-3 root subsystems of E8
T31  <- A6    the reduction averages over a reflection subgroup
T31  <- T10   A1+A1+A1 hosts no closed triangular history
T31  <- T25   A2+A1 is reducible, which is two operators not one
T31  <- T29   a rank-5 survivor means a rank-3 decayed part
T31  <- T30   exactly three candidates
T32  <- D14   every step is a primitive crossing
T32  <- T8    the 240 roots
T33  <- T5    the three weights exhaust the whole
T33  <- T24   the square of a defect has three terms and no others
T34  <- T16   one closure, so every traversal returns
T34  <- T27   one world, so simply connected
T34  <- T29   three spatial modes
T35  <- T8    the carrier equals its own dual, so one sphere not two
T35  <- T34   bound states are harmonics on the three-sphere
T36  <- T15   the record accumulates along one monotone order
```

```
T36  <- T25   matter closes into a cycle reproducing its readable identity
T36  <- T26   retention over replacement forbids the reverse
T37  <- A6    bimodularity makes the image an algebra
T37  <- T13   the readable object is the physical algebra, not a subspace
T38  <- T31   the three-two split of the surviving block
T39  <- T32   one retention per root over a complete closure
T39  <- T33   the cosmological term of the same square
O1   <- D15   A concrete representation of the cluster projection C
O2   <- T17   the surviving local symmetry must be the one T17 derives
O2   <- O1    OPEN, conditional on O1
O3   <- O1    OPEN, conditional on O1
O4   <- D14   a complete carrier closure
O4   <- T8    passes each of the 240 roots exactly once
O5   <- D11   the growth of the record
O7   <- T6    quantised retention at powers of r
N_lambda <- T8    (4 pi / sqrt3) r^240
N_lambda <- O4    The global vacuum residue rests on it
```

The graph is acyclic. This is the strongest structural statement the corpus can make about itself, and it is worth being exact about what it does and does not establish. It establishes that no claim in the framework is used in its own derivation. It does not establish that the claims are true, and it does not establish that the roots are the right roots. It rules out one specific failure, circularity, completely and checkably. The remaining ways to be wrong are in Part Seven and in Paper 21 Section Twelve.

PART SEVEN

Proof obligations

What the framework owes, with the reason each item stands. The corpus does not object to open questions; it objects to questions left open without anyone having looked. Every item below has been attacked, and five of the seven moved in this printing.

Before the list, one structural fact, because it changes how the list should be read. No result from T1 to T32 depends on any obligation. The obligations are terminal: nothing in the argument hangs off them. The single thing in the corpus resting on an open item is the standing vacuum readout, which rested on O4 and no longer does, since T32 proves the closure exists. The argument is therefore closed, and what follows is a forward work programme rather than a list of holes. That is checkable from the edge ledger in Part Six and is certified by the graph script.

Discharged, with what discharged them

O1 That the admissible reduction is an average over a reflection subgroup. DISCHARGED by T37. It was carried as a reading through four papers and then removed by a result the corpus had proved two printings earlier for an unrelated purpose: T13 requires an algebra rather than a subspace, so a rootless decayed part cannot supply a rank-five reductive gauge algebra. No rootless candidate sampled leaves more than twenty roots orthogonal to it, against the forty D5 needs.

O2 That the surviving spatial modes number three. DISCHARGED by T29, T30 and T31, and unconditional since T37. It is the same fact as the surviving symmetry being so(10): either both or neither.

O3 The gauge algebra, matter modules, chirality, families and anomaly freedom. DISCHARGED for content by T31, T36 and T38. Gauge so(10), matter in the spinor sixteen, three families, every Standard Model hypercharge and electric charge in place, anomalies verified by summation over the states actually built, and a chirality mechanism. What is NOT discharged is the Yukawa structure; see the exposure below.

O4 That a complete carrier closure passes each root exactly once. DISCHARGED for existence by T32, with an explicit witness verified independently of the search that found it. Uniqueness and the count remain open, finite, and not required by anything the corpus claims.

Standing

O5 The integral joining record growth to the measured expansion history. The zeroth-order clause is PROVED in Paper 19, and the vacuum readout implies a dark energy fraction and an age of the universe that bracket the measured values with nothing fitted. OPEN as a derivation of the residual from first principles. A consistency check passed is not a derivation and is not counted as one.

O6 The audited closure counts of the measurement pipelines. OPEN and empirical. The arithmetic ships and the Hubble stake stands at 0.55 standard deviations. Paper 24 extends the same audit to the dark energy comparison: the framework reads apparent evolution as what summing pipelines of different closure count produces, and its falsifier is a single-deed pipeline alone returning an equation of state away from minus one.

O7 A running, self-holding system returning quantised retention at powers of r under a convention declared before the count. OPEN and empirical. The framework's own stated condition of defeat. It cannot be settled at a desk, it has never been attempted, and it is not regraded.

The exposure

One result of the recent papers is not a success and is recorded here at the same volume as the successes, because a register that graded findings by how much it liked them would be worth nothing. T25 makes mass the failure of an operation to commute with the decay, and every root in the matter sector returns the same failure. The construction therefore makes the three families degenerate, which is not what the world does. The hierarchy must come from the Yukawa structure, which the construction does not reach. Three families is a real result; three degenerate families is a wrong prediction until the hierarchy is supplied.

One coupling, one Yukawa structure, one closure count, and two measurements. That is the complete list. It is shorter than the list this printing opened with because four obligations were discharged rather than deferred, and everything else in the chain from A1 to T39 is proved, conditional on a condition written into its own statement, or dissolved as a unit convention.

PART EIGHT

Where each result lives

Every numbered result, and the paper or movement carrying its full derivation.

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T1 T2 T3 T4 T5      Monograph, Movements Two, Four and Five
T6 T27 T28          Paper 18, The Quantised Crossing
T7                  Monograph, Movement Seven; exponent from T10
T8 T9               Monograph, Movement Nine; Papers 08 and 09
T10 T11             Paper 20, The Forward Closure, Section Four
T12                 Reproducibility, e8_golden_tests.py; Paper 09
T13 T14             Paper 20, Sections Five and Appendix; Movement Ten
T15 T16             Monograph, Movement Eleven
T17                 Monograph, Movement Twelve; Papers 16 and 17
T18                 Paper 19, The Closing Ledger, Section Two
T19 T20             Monograph, Movement Seventeen; Paper 13
T21 T22 T23         Monograph, Movement Eighteen; Papers 13 and 15
T24                 Monograph, Movement Nineteen; Paper 20, Section Nine
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T25	Monograph, Movement Twenty; Paper 20, Section Ten
T26	Monograph, Movement Twenty-Six
T29 T30 T31 T32	Paper 21, The Rank-Three Decay

Papers 04 05	the realisation principle and the snowflake closure theorem
Papers 08 09 10	the self-dual carrier, the golden skeleton, the root three
Papers 11 12	the interface closure and its proof ledger; historical
Papers 13 to 17	the semantic layer and the logic of physical action
Papers 18 19	the exact crossing price and the closing ledger
Papers 20 21	the forward closure and the rank-three decay

The Monograph carries the argument as a story and grades every claim in place. The Observational Standard carries what may honestly be said about a measured system. Systems of Interpretation carries the record of the framework being thrown at real systems, including its misses, which fall in the regimes its own grammar forbids, and which are retained deliberately: a framework that reported only its successes would be worth less. The Reproducibility layer carries the data, scripts and certificates, and every number in Part Five and every result in Paper 21 regenerates from it by one command. This volume carries none of that and duplicates none of it. It carries the shape.

PART NINE

Concordance of movement numbers

The Fourth Printing inserted four movements into the Monograph, so every movement from the tenth onward carries a number one to four higher than it did in the Third. Papers 18 and 19 were issued against the earlier numbering and are preserved as issued rather than silently edited, in keeping with the corpus discipline that corrections belong in new printings rather than in altered files. A reader following a cross-reference from either of those papers should read it through this table. Papers 20 and 21 use the current numbering throughout, and the Fifth Printing changes no movement numbers.

Third	Fourth/Fifth	Movement	
-	10	The descent	(new in Fourth)
-	11	What survives	(new in Fourth)
-	12	The shape of the world	(new in Fourth)
10	13	The record of the whole	
11	14	The equation as a finished journey	
12	15	The action and its double	
13	16	The container	
14	17	The unresolved world	
15	18	The bookkeeping of chance	
16	19	The carrying of sameness	
-	20	The one operator	(new in Fourth)
17	21	The extended traversal	
18	22	The watcher's machinery	
19	23	The ocean	
20	24	The architecture of one world	
21	25	The luminous wound	
22	26	The wound that could not forget	
23	27	What it claims, and what it leaves open	